UCS553: Enterprise Web Application

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Lab Experiment 1

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1. Write Java Program to Check Leap Year

LeapYear.java

Share Run

```java
import java.util.Scanner;

public class LeapYear {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter year: ");
        int year = sc.nextInt();

        if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0)) {
            System.out.println("Leap Year");
        } else {
            System.out.println("Not a Leap Year");
        }
    }
}
```

Output

```
Enter year: 2026
Not a Leap Year

=== Code Execution Successful ===
```

2. Write Java Program to Check Whether a Character is Alphabet or Not

AlphabetCheck.java

Share Run

```java
import java.util.Scanner;

public class AlphabetCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
            System.out.println("It is an Alphabet");
        } else {
            System.out.println("It is not an Alphabet");
        }
    }
}
```

Output

```
Enter a character: a
It is an Alphabet

=== Code Execution Successful ===
```

3. Write Java Program to Find Factorial of a Number

Factorial.java

Share Run

```java
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        System.out.println("Factorial = " + fact);
    }
}
```

Output

```
Enter a number: 5
Factorial = 120

=== Code Execution Successful ===
```

4. Write Java Program to Display Fibonacci Series

Fibonacci.java

Share Run Output

```java
import java.util.Scanner;

public class Fibonacci {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of terms: ");
        int n = sc.nextInt();

        int a = 0, b = 1;

        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");

            int c = a + b;
            a = b;
            b = c;
        }
    }
}
```

```
Enter number of terms: 4
0 1 1 2
=== Code Execution Successful ===
```

5. Write Java Program to Find GCD of two Numbers

GCD.java

Share Run Output

```java
public class GCD {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int gcd = 1;

        for (int i = 1; i <= a && i <= b; i++) {
            if (a % i == 0 && b % i == 0) {
                gcd = i;
            }
        }

        System.out.println("GCD = " + gcd);
    }
}
```

```
Enter first number: 5
Enter second number: 15
GCD = 5

=== Code Execution Successful ===
```

6. Write Java Program to Find LCM of two Numbers

```java
public class LCM {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int lcm = 1;

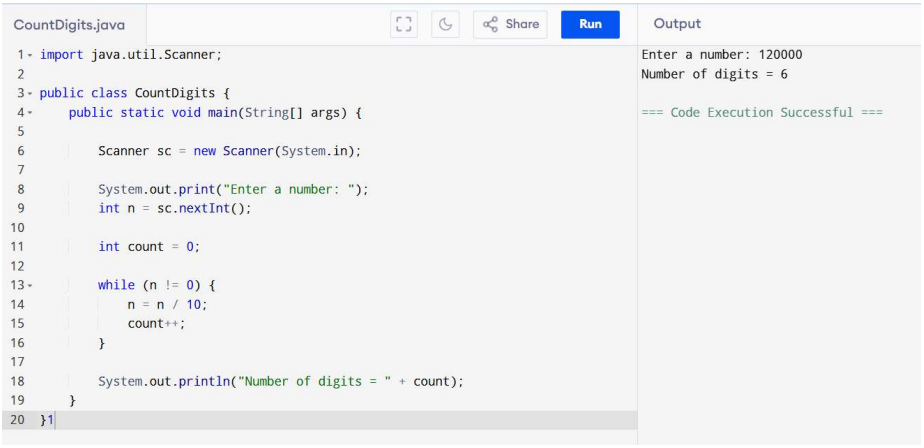
        for (int i = 1; i <= a * b; i++) {
            if (i % a == 0 && i % b == 0) {
                lcm = i;
                break;
            }
        }

        System.out.println("LCM = " + lcm);
    }
}
```

```
Enter first number: 125
Enter second number: 64
LCM = 8000

=== Code Execution Successful ===
```

7. Write Java Program to Count Number of Digits in an Integer



8. Write Java Program to Reverse a Number

ReverseNumber.java

Run

```java
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int rev = 0;

        while (n != 0) {
            int digit = n % 10;
            rev = rev * 10 + digit;
            n = n / 10;
        }

        System.out.println("Reverse = " + rev);
```

Output

```
Enter a number: 19
Reverse = 91

=== Code Execution Successful ===
```

9. Write Java Program to Calculate the Power of a Number

Power.java

Run

```java
import java.util.Scanner;

public class Power {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter base: ");
        int base = sc.nextInt();

        System.out.print("Enter exponent: ");
        int exponent = sc.nextInt();

        int result = 1;
        for (int i = 1; i <= exponent; i++) {
            result = result * base;
        }
        System.out.println("Power = " + result);
    }
}
```

Output

```
Enter a number: 19
Reverse = 91

=== Code Execution Successful ===
```

10. Write Java Program to Check Palindrome

Palindrome.java

Run

```java
import java.util.Scanner;

public class Palindrome {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int original = n;
        int rev = 0;

        while (n != 0) {
            int digit = n % 10;
            rev = rev * 10 + digit;
            n = n / 10;
        }

        if (original == rev) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not a Palindrome");
        }
```

Output

```
Enter a number: 727727
Palindrome

=== Code Execution Successful ===
```